

Land Use / Land Cover (LULC) Classification

Final Model: XGBoost with Hyperparameter Tuning

40 Multi-Seasonal Raw Spectral Features — Tuned XGBoost Classifier

Project Report

1. Model Overview

The final LULC classification model was developed using XGBoost with hyperparameter tuning. Based on the experimental results, the final model uses raw Sentinel-2 multispectral bands collected across four seasons (Q1–Q4).

The final feature representation contains 40 spectral features per sample, corresponding to 10 Sentinel-2 bands observed across four seasonal periods.

The final model was selected because it achieved the best overall performance among the evaluated configurations.

2. Final Dataset Configuration

Parameter	Configuration
Task	Multi-class LULC Classification
Dataset Size	90,000 samples
Number of Classes	9
Samples per Class	10,000
Train/Test Split	80/20
Training Samples	72,000
Test Samples	18,000
Temporal Coverage	4 Seasons (Q1–Q4)
Spectral Bands per Season	10
Total Input Features	40
Spectral Indices	None
Model	XGBoost Classifier
Hyperparameter Tuning	Yes
Random State	42

3. Input Feature Configuration

The final model uses the following Sentinel-2 spectral bands: B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12.

These bands are available for four seasonal observations: Q1, Q2, Q3, Q4.

Therefore: 10 spectral bands × 4 seasons = 40 input features.

The final model intentionally uses raw spectral bands only. Spectral indices were not included in the final feature set.

4. Class Distribution

The final dataset contains nine LULC classes with an equal number of samples.

Class Label	Samples
10	10,000
20	10,000
30	10,000
40	10,000
50	10,000
60	10,000
80	10,000
90	10,000
95	10,000
Total	90,000

The balanced class distribution ensures that no individual class dominates the training dataset.

5. XGBoost Model Configuration

The final XGBoost classifier was configured using the optimized hyperparameters obtained during the tuning stage.

Hyperparameter	Final Value
n_estimators	700
max_depth	10
learning_rate	0.05
subsample	0.8
colsample_bytree	0.8
objective	multi:softmax
num_class	9
eval_metric	mlogloss
random_state	42
n_jobs	-1

Hyperparameter Interpretation

- 700 estimators: The model builds 700 boosting trees to progressively improve predictions.
- Max depth = 10: Allows individual trees to capture relatively complex relationships between spectral features.
- Learning rate = 0.05: Uses smaller incremental updates to improve generalization.
- Subsample = 0.8: Each boosting iteration uses 80% of the training samples.
- Column sampling = 0.8: 80% of the features are considered for each tree.
- Multi:softmax: Directly produces one of the nine class labels.
- ML log-loss: Used as the evaluation metric during training.

6. Final Model Performance

The final model was evaluated on 18,000 previously unseen test samples.

Metric	Result
Accuracy	0.89 (~89%)
Macro Precision	0.89
Macro Recall	0.89
Macro F1-score	0.89
Weighted Precision	0.89
Weighted Recall	0.89
Weighted F1-score	0.89
Test Samples	18,000

Final Performance

- Accuracy $\approx$ 89%
- Macro F1-score = 0.89

The model correctly classified approximately 89% of the 18,000 test samples.

7. Class-wise Performance

Class	Precision	Recall	F1-score	Support
0	0.84	0.82	0.83	2,000
1	0.85	0.87	0.86	2,000
2	0.82	0.81	0.82	2,000
3	0.93	0.91	0.92	2,000

Class	Precision	Recall	F1-score	Support
4	0.85	0.87	0.86	2,000
5	0.88	0.89	0.88	2,000
6	0.99	0.98	0.99	2,000
7	0.86	0.89	0.87	2,000
8	0.96	0.92	0.94	2,000
Macro Avg	0.89	0.89	0.89	18,000
Weighted Avg	0.89	0.89	0.89	18,000

8. Class-wise Analysis

Best-performing class

Class 6 achieved the highest performance: Precision 0.99, Recall 0.98, F1-score 0.99.

This indicates that Class 6 has highly distinguishable spectral characteristics within the selected feature space.

Second-best class

Class 8 achieved: Precision 0.96, Recall 0.92, F1-score 0.94.

Strong-performing class

Class 3 achieved an F1-score of 0.92, also demonstrating strong classification performance.

Most challenging class

Class 2 achieved the lowest F1-score of 0.82. This suggests that Class 2 has greater spectral similarity with other LULC categories, making it more difficult for the model to distinguish.

9. Effect of Hyperparameter Tuning

Before hyperparameter tuning, the best XGBoost configuration from the multi-seasonal experiment achieved:

Model	Accuracy	Macro F1
XGBoost — Before Tuning	88.21%	0.88
XGBoost — Tuned	~89%	0.89

The tuned model therefore provided an improvement of approximately +0.7 percentage points in accuracy and +0.01 Macro F1.

Although the improvement was relatively small, it was obtained from an already strong baseline, indicating that the hyperparameter tuning provided a measurable refinement rather than a dramatic change.

10. Why This Model Was Selected

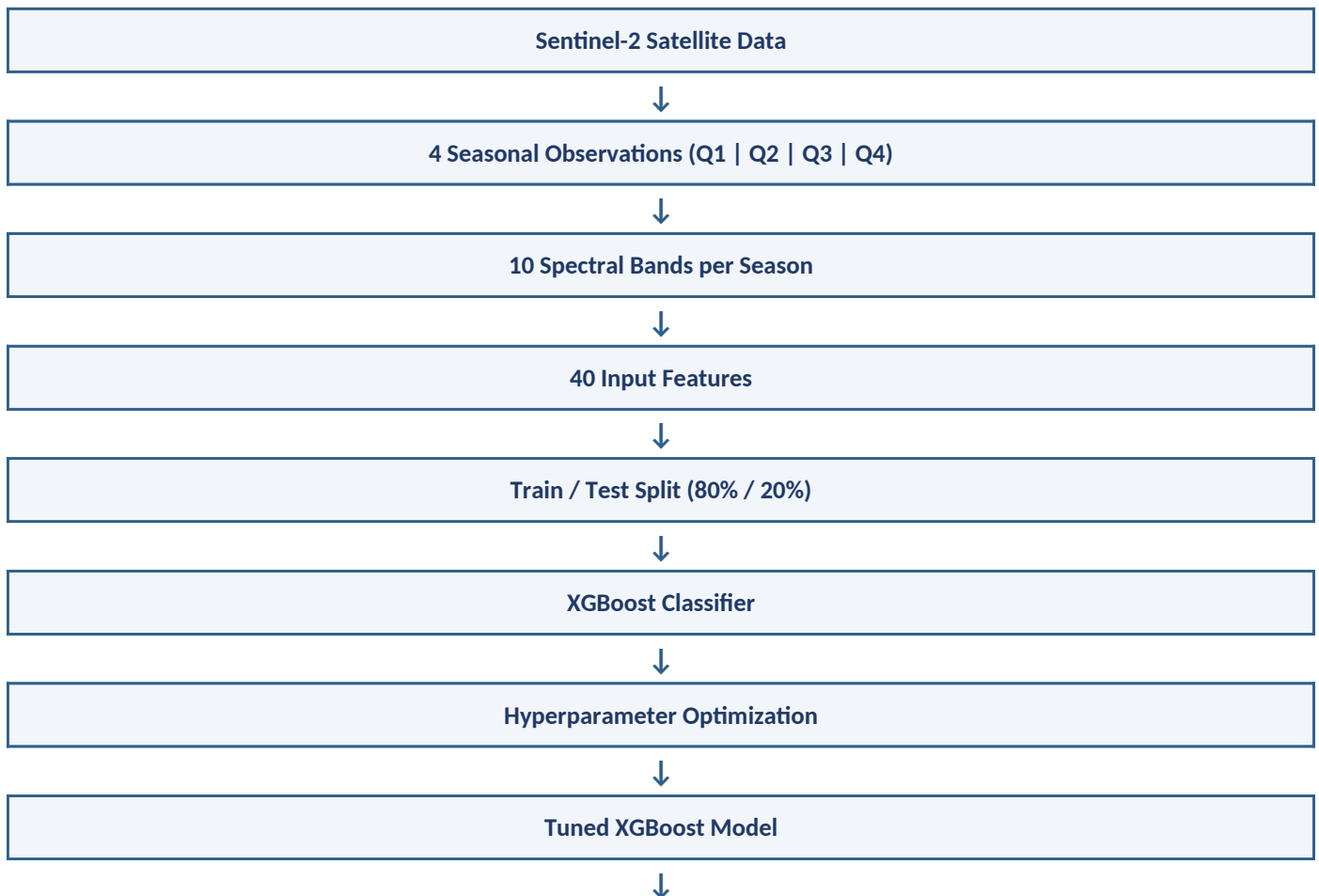
The final model was selected based on the complete experimental evaluation. The experiments demonstrated that:

- Multi-seasonal spectral information performed better than the single-season configuration.
- The 90,000-sample dataset provided a large and balanced training set.
- XGBoost generally outperformed Random Forest in the stronger multi-season configurations.
- Adding spectral indices did not improve Random Forest's overall performance in the tested configuration.
- The 40 raw multi-seasonal features provided a strong and relatively compact representation.
- Hyperparameter tuning further improved the XGBoost model.

Therefore, the final configuration was selected as: 40 multi-seasonal raw Sentinel-2 spectral features + tuned XGBoost.

11. Final Model Architecture

The complete final pipeline can be represented as:



9-Class LULC Prediction

12. Final Model Summary

Component	Final Selection
Input Data	Sentinel-2 multispectral data
Temporal Representation	4 seasons
Spectral Bands	10 per season
Total Features	40
Spectral Indices	Not used
Dataset Size	90,000
Classes	9
Samples/Class	10,000
Training Samples	72,000
Test Samples	18,000
Algorithm	XGBoost
Hyperparameter Tuning	Yes
Best Accuracy	~89%
Macro F1	0.89

13. Final Conclusion

The final LULC classification model was developed using 40 raw Sentinel-2 spectral features collected across four seasons and an optimized XGBoost classifier. The model was trained on a balanced dataset of 90,000 samples representing nine LULC classes, with 10,000 samples per class.

After hyperparameter tuning, the model achieved approximately 89% accuracy and a Macro F1-score of 0.89 on 18,000 unseen test samples. The model demonstrated strong class-wise performance, with F1-scores ranging from 0.82 to 0.99.

Class 6 achieved the highest F1-score of 0.99, while Class 2 was comparatively more challenging with an F1-score of 0.82. Overall, the results demonstrate that combining multi-seasonal Sentinel-2 spectral information with a tuned XGBoost classifier provides a strong and effective solution for nine-class LULC classification.